universe through the substance of space? If we are fundamentally unable to detect space, or changes within space, how can we claim it actually exists?

It is here that Newton stepped in with his bucket and dramatically changed the character of the debate. While Newton agreed that certain features of absolute space seem difficult or perhaps impossible to detect directly, he argued that the existence of absolute space does have consequences that are observable: accelerations, such as those at play in the rotating bucket, are accelerations with respect to absolute space. Thus, the concave shape of the water, according to Newton, is a consequence of the existence of absolute space. And Newton argued that once one has any solid evidence for something's existence, no matter how indirect, that ends the discussion. In one clever stroke, Newton shifted the debate about space from philosophical ponderings to scientifically verifiable data. The effect was palpable. In due course, Leibniz was forced to admit, "I grant there is a difference between absolute true motion of a body and a mere relative change of its situation with respect to another body."[8] This was not a capitulation to Newton's absolute space, but it was a strong blow to the firm relationist position.

During the next two hundred years, the arguments of Leibniz and others against assigning space an independent reality generated hardly an echo in the scientific community.[9] Instead, the pendulum had clearly swung to Newton's view of space; his laws of motion, founded on his concept of absolute space, took center stage. Certainly, the success of these laws in describing observations was the essential reason for their acceptance. It's striking to note, however, that Newton himself viewed all of his achievements in physics as merely forming the solid foundation to support what he considered his really important discovery: absolute space. For Newton, it was all about space.[10]

Mach and the Meaning of Space

When I was growing up, I used to play a game with my father as we walked down the streets of Manhattan. One of us would look around, secretly fix on something that was happening—a bus rushing by, a pigeon landing on a windowsill, a man accidentally dropping a coin—